

10. A* ALGORITHM

```
graph = {}

n = int(input("Enter number of nodes: "))

for i in range(n):
    node = input("Node: ")
    graph[node] = {}
    e = int(input("Edges from " + node + ": "))
    for j in range(e):
        v = input("Connected node: ")
        c = int(input("Cost: "))
        graph[node][v] = c

h = {}

for i in graph:
    h[i] = int(input("Heuristic of " + i + ": "))

start = input("Start node: ")
goal = input("Goal node: ")

open = [start]
cost = {start: 0}
parent = {start: None}

while open:
    x = min(open, key=lambda i: cost[i] + h[i])
    open.remove(x)
    if x == goal:
        path = []
        while x:
            path.append(x)
```

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        x = parent[x]
    print("Path:", path[::-1])
    break
for y in graph[x]:
    g = cost[x] + graph[x][y]
    if y not in cost or g < cost[y]:
        cost[y] = g
        parent[y] = x
    open.append(y)

```

OUTPUT:

```

Enter number of nodes: 4
Node: A
Edges from A: 2
Connected node: B
Cost: 1
Connected node: C
Cost: 4
Node: B
Edges from B: 1
Connected node: D
Cost: 2
Node: C
Edges from C: 1
Connected node: D
Cost: 1
Node: D
Edges from D: 0
Heuristic of A: 3
Heuristic of B: 2
Heuristic of C: 1
Heuristic of D: 0
Start node: A
Goal node: D
Path: ['A', 'B', 'D']

```